

MH1100 Calculus I

Handout 3 - Limits II

Limit Laws  can be proved using the ϵ - δ definition of limit

- Sum Law, Difference Law, Constant Multiple Law, Product Law, Quotient Law
- Direct Substitution Property, the Replacement Law
- The Squeeze Theorem

Definition  $\rightarrow$ Theorem $\rightarrow$ Application
To use, need to prove using definition

Limit laws

We will see later on that to actually use the definition of limit to prove limit equations can be difficult, especially when the functions are complicated.

What we normally do, however, is use theorems called **Limit Laws** to build up more complicated limits from simpler pieces.

↗ applicable to one-sided limits

Theorem (Sum law)

Let f and g be functions and let $a \in \mathbb{R}$. If the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

If not satisfied, it doesn't mean the sum function does not have a limit.

exist, then $\lim_{x \rightarrow a} (f(x) + g(x))$ exists and

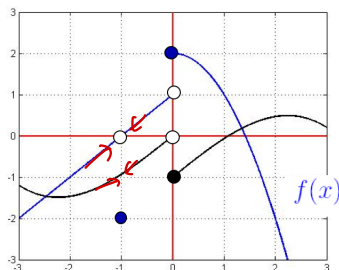
$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x).$$

That is to say, **the limit of a sum is the sum of the limits.**

Limit laws

Example A

Consider functions f and g with graphs as shown below. What does the limit law for sums have to say about $\lim_{x \rightarrow -1} (f(x) + g(x))$?



$$\lim_{x \rightarrow -1} f(x) = 0$$

$$\lim_{x \rightarrow -1} g(x) = -1$$

Solution:

Notice that the limits $\lim_{x \rightarrow -1} f(x)$ and $\lim_{x \rightarrow -1} g(x)$ both exist. Thus, the assumptions of the sum law are satisfied. The law then concludes that $\lim_{x \rightarrow -1} (f(x) + g(x))$ exists and $\lim_{x \rightarrow -1} (f(x) + g(x)) = -1$.

Limit laws

$$\lim_{x \rightarrow 0^+} [f(x) + g(x)] = \lim_{x \rightarrow 0^+} f(x) + \lim_{x \rightarrow 0^+} g(x) = 2 + (-1) = 1$$

Exercise

$$\lim_{x \rightarrow 0^-} [f(x) + g(x)] = \lim_{x \rightarrow 0^-} f(x) + \lim_{x \rightarrow 0^-} g(x) = 1 \Rightarrow \text{sum law works for one sided limit}$$

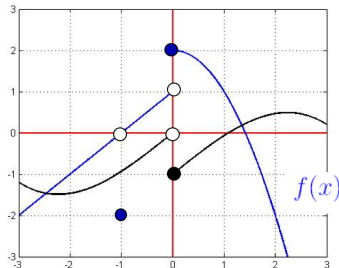
$$\lim_{x \rightarrow 0} [f(x) + g(x)] = 1$$

Since the one sided limits are equal
 $\rightarrow$ this conclusion is not from the sum law.

Consider functions f and g with graphs as shown below. What does the limit law for sums have to say about $\lim_{x \rightarrow 0} (f(x) + g(x))$?

* cannot directly use the sum law.

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= 1 \\ \lim_{x \rightarrow 0^+} f(x) &= 2 \\ \therefore \lim_{x \rightarrow 0} f(x) &\text{ does not exist} \\ \lim_{x \rightarrow 0^-} g(x) &= 0 \\ \lim_{x \rightarrow 0^+} g(x) &= -1 \\ \therefore \lim_{x \rightarrow 0} g(x) &\text{ does not exist} \end{aligned}$$



Solution:

Notice that neither of the limits $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ exist. Thus, the assumptions of the sum law are **NOT** satisfied. So the sum law has nothing to say about this limit.

Theorem (Product law)

Let f and g be functions and let $a \in \mathbb{R}$. If the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist, then

$$\lim_{x \rightarrow a} f(x)g(x)$$

exists and

$$\lim_{x \rightarrow a} f(x)g(x) = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x).$$

That is to say, the limit of a product is the product of the limits.

Theorem (Quotient law)

Let f and g be functions and let $a \in \mathbb{R}$. If the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist, and

$$\lim_{x \rightarrow a} g(x) \neq 0, \quad \rightarrow \because \text{denominator} \neq 0$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

exists and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}.$$

That is to say, the limit of a quotient is the quotient of the limits.

There are also two simple functions whose limits we can assume. (Later we will prove these limits.)

Theorem (Limits of simple functions)

Let $a \in \mathbb{R}$, and $c \in \mathbb{R}$.

•

$$\lim_{x \rightarrow a} c = c$$

constant function $\Rightarrow$ it's just a straight line.

•

$$\lim_{x \rightarrow a} x = a$$

Limit laws

The following limit law is a consequence of the other limit laws.

Exercise: Explain why.

can be derived from sum law and product law, as well as the two on the previous slide.

Theorem (Difference law)

Let f and g be functions and let $a \in \mathbb{R}$. If the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist, then $\lim_{x \rightarrow a} (f(x) - g(x))$ exists and

$$\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x).$$

That is to say, *the limit of a difference is the difference of the limits.*

$f(x) - g(x) = f(x) + (-1)g(x)$

$\lim_{x \rightarrow a} (-1) = -1$	$\left. \begin{array}{l} \text{product law} \\ \lim_{x \rightarrow a} g(x) \text{ exists} \end{array} \right\} \Rightarrow \lim_{x \rightarrow a} (-1)g(x) = (-1) \lim_{x \rightarrow a} g(x)$	$\lim_{x \rightarrow a} f(x) \text{ exists}$	$\left. \begin{array}{l} \text{sum law} \\ \lim_{x \rightarrow a} g(x) \text{ exists} \end{array} \right\} \Rightarrow \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} (-1)g(x) = \lim_{x \rightarrow a} f(x) + (-1) \lim_{x \rightarrow a} g(x)$
		$\lim_{x \rightarrow a} g(x) \text{ exists}$	

Handwritten notes in blue and red ink showing the derivation of the difference law from the sum and product laws.

→ can be proved using product law.

Theorem (Constant multiple law)

Let f be a function and let $a \in \mathbb{R}$. Suppose that c is a constant. If the limit

$$\lim_{x \rightarrow a} f(x)$$

exists, then

$g(x) = c$
a constant function.

$$\lim_{x \rightarrow a} cf(x)$$

exists and

$$\lim_{x \rightarrow a} cf(x) = c \lim_{x \rightarrow a} f(x).$$

That is to say, the limit of a constant times a function is the constant times the limit of the function.

Limit laws

If we use the Product Law repeatedly with $g(x) = f(x)$, we obtain the following Power Law.

Theorem (Power law)

Let f be a function and let $a \in \mathbb{R}$. Suppose that n is a positive integer. If the limit

$$\lim_{x \rightarrow a} f(x)$$

$[f(x)]^n = f(x) \cdot g(x)$
where $g(x) = f(x)^{n-1}$

exists, then

$$\lim_{x \rightarrow a} [f(x)]^n$$

exists and

$$\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n.$$

Limit laws

Here is another useful limit law. This is the last one we will meet for a while.

Theorem (Root law)

Let f be a function and let $a \in \mathbb{R}$. Suppose that n is a positive integer. If the limit

$$\lim_{x \rightarrow a} f(x)$$

exists, then

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)}$$

exists and

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}.$$

If n is even, we assume that $\lim_{x \rightarrow a} f(x) > 0$.

When n is even
if $\lim_{x \rightarrow a} f(x) = 0$
 $\sqrt[n]{f(x)}$ may not be defined
when $x \rightarrow a$

Example C

Does the following limit exist, and, if it does, evaluate it

$$\lim_{x \rightarrow 3} (x^2 - 9)$$

A quick and casual solution:

$$\begin{aligned}\lim_{x \rightarrow 3} (x^2 - 9) &= \lim_{x \rightarrow 3} x^2 + \lim_{x \rightarrow 3} (-9) && \text{by the Sum Law} \\ &= \lim_{x \rightarrow 3} x \cdot \lim_{x \rightarrow 3} x + \lim_{x \rightarrow 3} (-9) && \text{by the Product Law} \\ &= 3 \times 3 + (-9) && \text{Limits of simple functions} \\ &= 0.\end{aligned}$$

At each step, we assume the involved limits exist.

Therefore, the limit $\lim_{x \rightarrow 3} (x^2 - 9)$ exists and

$$\lim_{x \rightarrow 3} (x^2 - 9) = 0.$$

Example C–Solution

A more rigorous solution:

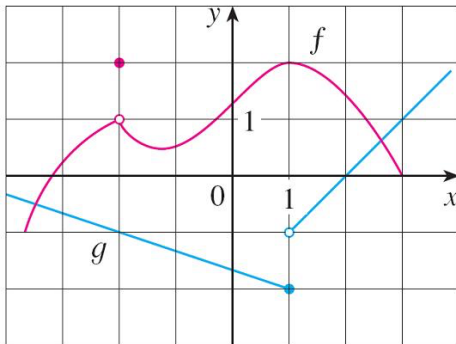
- To begin, note that $\lim_{x \rightarrow 3} x$ exists and equals 3. (by the limit law for simple functions)
- Thus, $\lim_{x \rightarrow 3} x^2$ exists and equals $(\lim_{x \rightarrow 3} x)(\lim_{x \rightarrow 3} x) = 9$. (by the limit law for products)
- Note that $\lim_{x \rightarrow 3} (-9)$ exists and equals -9 . (by the limit law for simple functions)
- Finally, we can apply the limit law for sums to deduce that, because the limits $\lim_{x \rightarrow 3} x^2$ and $\lim_{x \rightarrow 3} (-9)$ both exist (by the previous two deductions), the $\lim_{x \rightarrow 3} (x^2 - 9)$ does indeed exist and is equal to $\lim_{x \rightarrow 3} x^2 + \lim_{x \rightarrow 3} (-9)$, which equals 0. (by the limit law for sums, together with the previous two deductions.)

Limit laws

Example 1

Use the Limit Laws and the graphs of f and g in the figure below to evaluate the following limits, if they exist.

(a) $\lim_{x \rightarrow -2} [f(x) + 5g(x)]$ (b) $\lim_{x \rightarrow 1} [f(x)g(x)]$ (c) $\lim_{x \rightarrow 2} \frac{f(x)}{g(x)}$



Example 1(a)–Solution

Solution:

(a) From the graphs of f and g we see that

$$\lim_{x \rightarrow -2} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow -2} g(x) = -1$$

Therefore we have

$$\begin{aligned} \lim_{x \rightarrow -2} [f(x) + 5g(x)] &= \lim_{x \rightarrow -2} f(x) + \lim_{x \rightarrow -2} [5g(x)] && \text{by the Sum Law} \\ &= \lim_{x \rightarrow -2} f(x) + 5 \lim_{x \rightarrow -2} g(x) && \text{Constant Multiple Law} \\ &= 1 + 5 \times (-1) \\ &= -4 \end{aligned}$$

Hence,

$$\lim_{x \rightarrow -2} [f(x) + 5g(x)] = -4$$

Example 1(b)–Solution

Comment: When computing one-sided limits, we have the fact that the Limit Laws also hold for one-sided limits.

Solution:

(b) We see that $\lim_{x \rightarrow 1} f(x) = 2$. But $\lim_{x \rightarrow 1} g(x)$ does not exist because the left and right limits are different:

$$\lim_{x \rightarrow 1^-} g(x) = -2 \quad \text{and} \quad \lim_{x \rightarrow 1^+} g(x) = -1$$

So we cannot use the Product Law for the desired limit.

But we can use the Product Law for the one-sided limits:

$$\lim_{x \rightarrow 1^-} [f(x)g(x)] = \lim_{x \rightarrow 1^-} f(x) \cdot \lim_{x \rightarrow 1^-} g(x) = 2 \cdot (-2) = -4$$

$$\lim_{x \rightarrow 1^+} [f(x)g(x)] = \lim_{x \rightarrow 1^+} f(x) \cdot \lim_{x \rightarrow 1^+} g(x) = 2 \cdot (-1) = -2$$

The left and right limits are NOT equal, so $\lim_{x \rightarrow 1} [f(x)g(x)]$ does not exist.
 ↳ not defined.

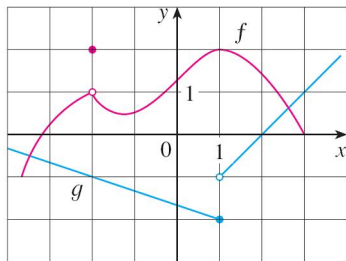
Example 1(c)–Solution

Solution:

(c) The graphs show that

$$\lim_{x \rightarrow 2} f(x) \approx 1.4 \quad \text{and} \quad \lim_{x \rightarrow 2} g(x) = 0$$

Because the limit of the denominator is 0, we cannot use the Quotient Law.



*infinite limit
→ not defined
⇒ does not exist.*

It is obvious that the given limit **does not exist** because the denominator approaches 0 while the numerator approaches a nonzero number.

Limit laws

Example 2

Evaluate the following limits and justify each step.

(a) $\lim_{x \rightarrow 5} (2x^2 - 3x + 4)$ (b) $\lim_{x \rightarrow 5} \frac{2x^2 - 3x + 4}{5 - 3x}$

Solution:

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow 5} (2x^2 - 3x + 4) &= \lim_{x \rightarrow 5} (2x^2 - 3x) + \lim_{x \rightarrow 5} 4 = \lim_{x \rightarrow 5} 2x^2 - \lim_{x \rightarrow 5} 3x + 4 \\ &= 2 \lim_{x \rightarrow 5} x^2 - 3 \lim_{x \rightarrow 5} x + 4 \\ &= 2 \times (5^2) - 3 \times 5 + 4 \\ &= 39 \end{aligned}$$

Handwritten annotations:

- sum law* (pointing to $\lim_{x \rightarrow 5} (2x^2 - 3x) + \lim_{x \rightarrow 5} 4$)
- simple fact* (pointing to $\lim_{x \rightarrow 5} 4 = 4$)
- difference law* (pointing to $\lim_{x \rightarrow 5} 2x^2 - \lim_{x \rightarrow 5} 3x + 4$)
- constant multiple* (pointing to $2 \lim_{x \rightarrow 5} x^2$)
- power law* (pointing to $\lim_{x \rightarrow 5} x^2 = 5^2$)
- simple fact* (pointing to $5^2 = 25$)

Question: Let $f(x) = 2x^2 - 3x + 4$. It is clear that f is defined at $x = 5$. What is the value of $f(5)$?

Example 2—Solution

Solution: (b) We have found the limit of the numerator in Part (a):

$$\lim_{x \rightarrow 5} (2x^2 - 3x + 4) = 39.$$

For the denominator $5 - 3x$, we can evaluate its limit at $x = 5$

$$\begin{aligned}\lim_{x \rightarrow 5} (5 - 3x) &= \lim_{x \rightarrow 5} 5 - \lim_{x \rightarrow 5} (3x) = 5 - 3 \lim_{x \rightarrow 5} x \\ &= 5 - 3 \times 5 = -10 \neq 0\end{aligned}$$

Therefore, using the Quotient Law, we have

$$\lim_{x \rightarrow 5} \frac{2x^2 - 3x + 4}{5 - 3x} = -\frac{39}{10} = -3.9.$$

Question: Let $g(x) = (2x^2 - 3x + 4)/(5 - 3x)$. It is obvious that g is defined at $x = 5$. What is the value of $g(5)$? $g(5) = -3.9$

Direct Substitution Property

If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

$f(a)$ is defined.
 $\rightarrow$ its denominator is non zero
if f is a rational function.

Functions with the Direct Substitution Property are called *continuous at a* and will be studied in Section 1.8. However, not all limits can be evaluated by direct substitution.

$$f(x) = a_0x^4 + a_1x^{n-1} + \dots + a_{n-1}x + a_n$$
$$\lim_{x \rightarrow a} f(x) = f(a)$$

Limit laws

* Cannot use L'Hopital's Rule for this question.

Example 3

Evaluate the limit $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

① L'Hopital's Rule is only for differential functions.
② A function is differentiable at a
 $\Rightarrow$ its derivative at $x=a$ is defined. $f'(a)$ is defined.
③ $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists.

Solution:

Note that $\lim_{x \rightarrow 1} (x - 1) = 0$.

This means we cannot just use the Quotient Law, and need to take an **alternative approach** to understand this limit. Instead,

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &\downarrow \\ &x \neq 1 \Rightarrow x - 1 \neq 0 \\ &\therefore \text{can cancel common factor.} \\ &= \lim_{x \rightarrow 1} (x + 1) \\ &= 1 + 1 = 2 \end{aligned}$$

Question: But are we really allowed to cancel like this?

Limit laws

So why were we allowed to cancel, exactly?

We were implicitly using a very simple but important fact which we will sometimes call the *Replacement Law*.

Theorem: The replacement law

Let f and g be functions and let $a \in \mathbb{R}$. Assume that $f(x) = g(x)$ whenever $x \neq a$. Then either:

- Both f and g have limits at a and

$$f(x) = \frac{x^2 - 1}{x - 1}$$

$$g(x) = x + 1$$

$$f(x) = g(x) \text{ when } x \neq 1$$

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x).$$

- Or, neither limit exists.

Comment: This law is a simple deduction from the fact that a limit $\lim_{x \rightarrow a} f(x)$ has nothing to do with the actual value of the function $f(x)$ at a .

Example 3 (Cont'd)

Let's look at the calculation again

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x + 1) \\ &= 1 + 1 = 2\end{aligned}$$

We are allowed to cancel $(x - 1)$ here because the functions $f(x) = (x^2 - 1)/(x - 1)$ and $g(x) = (x + 1)$ have **exactly the same values** at every point except at the point where we are taking the limit, $x = 1$. Thus we can **apply the replacement law**.

Comment: We do not have an infinite limit even though the denominator approaches 0 as $x \rightarrow 1$. When both numerator and denominator approach 0, the limit may be infinite or it may be some finite value.

Limit laws

Exercise (Eliminating Common Factors from Zero Denominators)

Evaluate the limit $\lim_{x \rightarrow -2} \frac{x^2+x-2}{x^2+2x}$.

We cannot apply the Quotient Law immediately, since the limit of the denominator is 0. If the denominator is zero, cancelling common factors in the numerator and denominator may reduce the fraction to one whose denominator is no longer zero at $x=0$. If this happens, we can find the limit by substitution in the simplified fraction.

$$\begin{aligned}\lim_{x \rightarrow -2} \frac{x^2+x-2}{x^2+2x} &= \lim_{x \rightarrow -2} \frac{(x+2)(x-1)}{x(x+2)} \\ &= \lim_{x \rightarrow -2} \frac{x-1}{x} \\ &= \frac{1}{2}\end{aligned}$$

Limit laws

Exercise (Eliminating Common Factors from Zero Denominators)

Evaluate the limit $\lim_{t \rightarrow 0} \frac{\sqrt{t^2+400}-20}{t^2}$.

We cannot apply the Quotient Law immediately, since the limit of the denominator is 0. But we can rationalize the numerator.

$$\begin{aligned} & \lim_{t \rightarrow 0} \frac{\sqrt{t^2+400}-20}{t^2} \\ &= \lim_{t \rightarrow 0} \frac{(\sqrt{t^2+400}-20)(\sqrt{t^2+400}+20)}{t^2(\sqrt{t^2+400}+20)} \\ &= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2+400}+20} \cdot \lim_{t \rightarrow 0} \frac{1}{t^2} \quad \text{Quotient Law} \\ &= \frac{\lim_{t \rightarrow 0} 1}{\lim_{t \rightarrow 0} (\sqrt{t^2+400}+20)} \\ &= \frac{1}{\lim_{t \rightarrow 0} \sqrt{t^2+400} + \lim_{t \rightarrow 0} 20} \\ &= \frac{1}{\sqrt{\lim_{t \rightarrow 0} (t^2+400)} + \lim_{t \rightarrow 0} 20} \\ &= \frac{1}{40} \end{aligned}$$

Limit laws

Some limits are best calculated by first finding the left- and right-hand limits. The following theorem says that a two-sided limit exists if and only if both of the one-sided limits exist and are equal.

Theorem 1

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x).$$

When computing one-sided limits, we use the fact that the Limit Laws also hold for one-sided limits.

Example 8

Prove that the following limit does not exist.

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

Example 8–Solution

Solution:

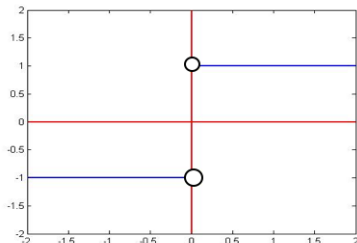
Using the facts that $|x| = x$ when $x > 0$ and $|x| = -x$ when $x < 0$, we have

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1$$

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} (-1) = -1$$

Since the right- and left-hand limits are different, it follows from Theorem 1 that

$\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.



$$\frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Limit laws

Exercise:

$$f(x) = |x| \quad \lim_{x \rightarrow 0} |x| = 0 = f(0)$$

$$\lim_{x \rightarrow 0} f(x) = f(0)$$

Evaluate the value of the limit $\lim_{x \rightarrow 0} |x|$

Recall that

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Since $|x| = x$ for $x > 0$, we have

$$\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$$

For $x < 0$, we have $|x| = -x$ and so

$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

Therefore, by Theorem 1,

$$\lim_{x \rightarrow 0} |x| = 0$$

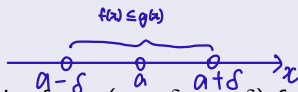
$$\begin{aligned} & \lim_{x \rightarrow 0^-} (-x) \\ &= (-1) \lim_{x \rightarrow 0^-} x \\ &= -1 \cdot 0 \\ &= 0 \end{aligned}$$

The next two theorems give two additional properties of limits.

Theorem 2

Let f and g be functions, and let $a \in \mathbb{R}$. Assume that

- $\lim_{x \rightarrow a} f(x)$ exists
- $\lim_{x \rightarrow a} g(x)$ exists
- $f(x) \leq g(x)$ for all x in an open interval of the form $(a - \delta, a + \delta)$ for some positive $\delta \in \mathbb{R}$ (except possibly at a itself).



Then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x).$$

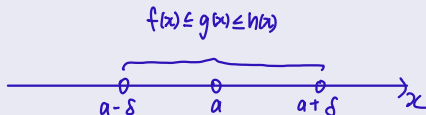
Limit laws

extension of theorem 2, can use theorem 2 to prove theorem 3

Theorem 3 (The squeeze theorem)

Let f , g , and h be functions, and let $a \in \mathbb{R}$. Assume that

- $\lim_{x \rightarrow a} f(x)$ exists
- $\lim_{x \rightarrow a} h(x)$ exists
- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$
- $f(x) \leq g(x) \leq h(x)$ for all x in an open interval of the form $(a - \delta, a + \delta)$ for some positive $\delta \in \mathbb{R}$ (except possibly at a itself).



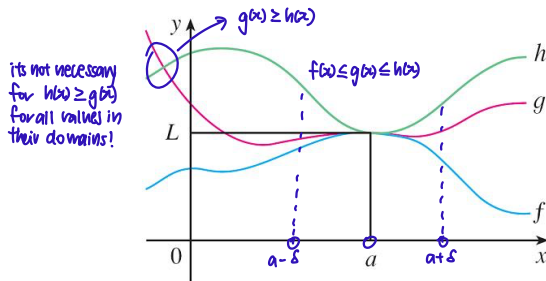
Then

- $\lim_{x \rightarrow a} g(x)$ exists
- $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$.

It's not necessary that $f(x) \leq g(x) \leq h(x)$ in their domains.
Only if there exists one open interval

Limit laws

The Squeeze Theorem, which is sometimes called the Sandwich Theorem or the Pinching Theorem.



It says that if $g(x)$ is squeezed between $f(x)$ and $h(x)$ near a , and if f and h have the same limit L at a , then g is forced to have the same limit L at a .

Limit laws

$$a \in \mathbb{R} : -|a| \leq a \leq |a|$$

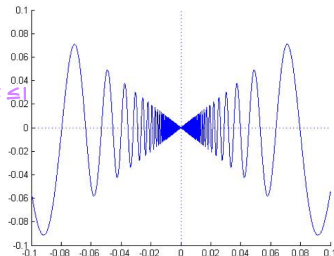
Example D

Evaluate the value of the limit $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$. **cannot use product law.
 $\lim_{x \rightarrow 0} \sin(x)$ does not exist.*

To use the squeeze theorem, we need to find out two other functions $f(x)$ and $h(x)$

$\therefore$ Use squeeze theorem.

$$\begin{aligned} -|x \sin x| &\leq x \sin x \leq |x \sin x| \\ \text{"always } \leq 1 \text{"} & \quad \text{"always } \leq 1 \text{"} \\ -|x| \cdot |\sin x| &\geq -|x| \quad |x| \cdot |\sin x| \leq |x| \\ \therefore -|x| &\leq x \sin x \leq |x| \end{aligned}$$



This plot leads us to expect that $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$. But this limit is not calculated by a combination of the various limit laws. Instead we need the squeeze theorem to determine the limit.

Example D—Solution

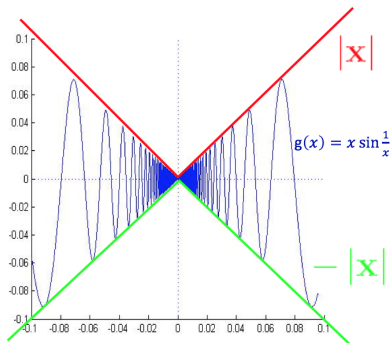
Now let's go back and apply the squeeze theorem to show that $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$.

Usually when we use the squeeze theorem we have two questions to answer:

(1) What should we set $f(x)$ to be?

(2) What should we set $h(x)$ to be?

Answers: $f(x) = -|x|$ and $h(x) = |x|$.



Example D—Solution (Cont'd)

To finish we have to show why the assumptions of the squeeze theorem are true for our choices of $f(x) = -|x|$ and $h(x) = |x|$.

Let's go through the 4 assumptions in turn.

(1) Why does $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (-|x|)$ exist?

Earlier we used 1-sided limits to find $\lim_{x \rightarrow 0} |x| = 0$. It follows that $\lim_{x \rightarrow 0} (-|x|)$ exists and equals 0.

(2) Why does $\lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} |x|$ exist?

Earlier we used 1-sided limits to find $\lim_{x \rightarrow 0} |x| = 0$.

(3) Why does $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} h(x)$?

Above we determined that both of these limits equals 0.

Example D—Solution (Cont'd)

(4) Why is $f(x) \leq g(x) \leq h(x)$?

There are two inequalities here. The second inequality is shown by the deduction:

$$\begin{aligned}g(x) &= x \sin \frac{1}{x} \leq |x \sin \frac{1}{x}| \\&= |x| |\sin \frac{1}{x}| \leq |x| = h(x)\end{aligned}$$

The other inequality is shown in a very similar way:

$$\begin{aligned}g(x) &= x \sin \frac{1}{x} \geq -|x \sin \frac{1}{x}| \\&= -|x| |\sin \frac{1}{x}| \geq -|x| = f(x)\end{aligned}$$

We have shown that all four assumptions of the squeeze theorem are satisfied, so we can conclude that $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ exists and equals 0.

Limit Laws

Example E

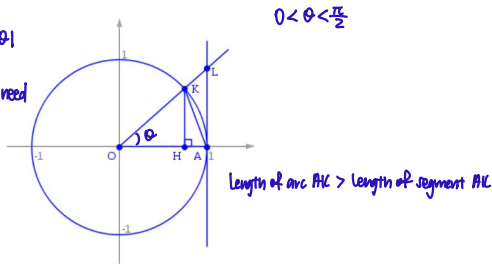
Prove that for any angle x measured in radians, the sine and cosine functions satisfy

$$-|\theta| \leq \sin \theta \leq |\theta| \quad \text{and} \quad -|\theta| \leq 1 - \cos \theta \leq |\theta|.$$

If $\theta > 1$, $\sin \theta \leq 1 < |\theta|$

If $\theta < -1$

$\Rightarrow$ automatically satisfied, no need to prove.

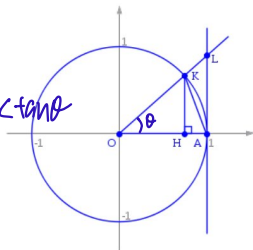


To establish these inequalities, we picture θ as a nonzero angle in standard position (see the figure above). The circle in the figure is a unit circle, so $|\theta|$ equals the **length of the circular arc AK**. The length of line segment AK is therefore not greater than $|\theta|$.

Example E—Solution (Cont'd)

Unit Circle

$$\left. \begin{array}{l} |HK| = \sin \theta \\ |\widehat{AK}| = \theta \\ |AL| = \tan \theta \end{array} \right\} \sin \theta < \theta < \tan \theta$$



$$\begin{aligned} \text{Area of } \triangle OKL &= \frac{1}{2} |OK| \cdot |HK| \\ &= \frac{1}{2} \sin \theta \end{aligned}$$

$$\text{Area of } \widehat{OK} = \frac{1}{2} r^2 \theta$$

$$\begin{aligned} \text{Area of } \triangle OAL &= \frac{1}{2} |OA| \cdot |AL| \\ &= \frac{1}{2} \tan \theta \end{aligned}$$

Triangle AKH is a right triangle with sides of length $|OH| = \cos \theta$

$$HK = |\sin \theta|, \quad AH = 1 - \cos \theta.$$

From the Pythagorean theorem and the fact that $AK \leq |\theta|$, we get

$$\sin^2 \theta + (1 - \cos \theta)^2 = (AK)^2 \leq \theta^2$$

$\rightarrow$ to include $\theta = 0$
 $\hookrightarrow$ since length of segment $AL <$ length of arc AK

The terms on the left-hand side of the above equation are both positive, so each is smaller than their sum and hence is less than or equal to θ^2 :

Example E—Solution (Cont'd)

From the Pythagorean theorem and the fact that $AK \leq |\theta|$, we get

$$\sin^2 \theta + (1 - \cos \theta)^2 = (AK)^2 \leq \theta^2.$$

The terms on the left-hand side of the above equation are both positive, so each is smaller than their sum and hence is less than or equal to θ^2 :

$$\sin^2 \theta \leq \theta^2 \quad \text{and} \quad (1 - \cos \theta)^2 \leq \theta^2.$$

By taking square roots, this is equivalent to saying that

$$|\sin \theta| \leq |\theta| \quad \text{and} \quad |1 - \cos \theta| \leq |\theta|,$$

so

squeeze theorem

$$-|\theta| \leq \sin \theta \leq |\theta| \quad \text{and} \quad -|\theta| \leq 1 - \cos \theta \leq |\theta|.$$

$$\begin{aligned} \lim_{\theta \rightarrow 0} (-|\theta|) &= 0 \\ \lim_{\theta \rightarrow 0} (|\theta|) &= 0 \end{aligned} \Rightarrow \lim_{\theta \rightarrow 0} \sin \theta = 0 \quad \rightarrow \text{similarly to find } \lim_{\theta \rightarrow 0} (1 - \cos \theta) = 0$$

Limit laws

Exercise:

The squeeze theorem helps us establish several important rules:

- $\lim_{x \rightarrow 0} \sin x = 0$
- $\lim_{x \rightarrow 0} \cos x = 1$
- For any function f , $\lim_{x \rightarrow a} |f(x)| = 0$ implies $\lim_{x \rightarrow a} f(x) = 0$.

$$\cos x = 1 - (1 - \cos x)$$

$$\lim_{x \rightarrow 0} (1 - \cos x) = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} \cos x = 1$$

$$\lim_{x \rightarrow 0} 1 = 1$$

↓
using difference
law

$$-|a| \leq a \leq |a|$$

$$-|f(x)| \leq f(x) \leq |f(x)| \quad \text{constant multiple law}$$

$$\lim_{x \rightarrow a} |f(x)| = 0 \Rightarrow \lim_{x \rightarrow a} (-|f(x)|) = 0$$

Limit laws

Exercise:

Use the squeeze theorem to prove that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

→ cannot use this limit to prove the limit

$$-|x| \leq \sin x \leq |x| \rightarrow \text{proven just now}$$

$$-1 \leq \frac{\sin x}{|x|} \leq 1$$

$\lim_{x \rightarrow 0^+} \frac{\sin x}{x} \rightarrow$ consider this case first

Since $\sin x < x < \tan x$

$$\sin x < x \Rightarrow \frac{\sin x}{x} < 1$$

$$x < \tan x = \frac{\sin x}{\cos x} \Rightarrow \cos x < \frac{\sin x}{x}$$

$$\therefore \cos x < \frac{\sin x}{x} < 1$$

$$\lim_{x \rightarrow 0} \cos x = 1 \quad \lim_{x \rightarrow 0} 1 = 1$$

Using squeeze theorem, $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$

$\frac{\sin x}{x}$ is an even function.

$$\therefore \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1 = \lim_{x \rightarrow 0^-} \frac{\sin x}{x}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

↓
the LHS is -1, we
cannot squeeze
 $\frac{\sin x}{|x|}$

$\sin x < x < \tan x \Rightarrow$ from slide 38